

Statistics Interview Questions

Entry-Level Data Analyst Interview Prep - Topics completed through Covariance & Correlation

How to use: Try answering each question aloud before reading the answer. Focus on understanding and interpretation first; formulas and coding come after the concept.

Topic area	Priority	Interview focus
Mean / Median / Mode	Very High	Concept + outliers + when to use which
Variance / Standard Deviation	Very High	Meaning, interpretation, n-1
Percentiles / Quartiles / IQR	Very High	Outliers + box plots
Histogram / Skewness	Very High	Interpret distributions
Covariance / Correlation	Very High	Direction, strength, causation
Population / Sample / Sampling	Very High	Scenario questions

1. Introduction to Statistics

Q1. What is Statistics?

Answer: Statistics is the process of collecting, organizing, analyzing, interpreting, and presenting data to understand patterns and make decisions.

Q2. Why is Statistics important for a Data Analyst?

Answer: Statistics helps a Data Analyst summarize data, identify patterns and outliers, measure relationships between variables, understand distributions, and make conclusions from data.

Q3. Is Statistics only about calculating averages?

Answer: No. Statistics also includes variability, distributions, relationships between variables, sampling, probability, and statistical inference.

Q4. What is the difference between data and statistics?

Answer: Data is raw information collected, while statistics are numerical measures or methods used to summarize and analyze that data.

2. Types of Statistics

Q5. What are the two main types of Statistics?

Answer: Descriptive Statistics summarizes the data you already have. Inferential Statistics uses sample data to draw conclusions about a larger population.

Q6. Give an example of descriptive statistics.

Answer: Calculating the average salary of employees in a company is descriptive statistics.

Q7. Give an example of inferential statistics.

Answer: Taking a sample of customers and using it to estimate the satisfaction level of all customers is inferential statistics.

Q8. Which type of Statistics is used when we calculate mean, median, and standard deviation?

Answer: These are examples of descriptive statistics.

3. Population and Sample

Q9. What is a population?

Answer: A population is the complete group of individuals or observations that we are interested in studying.

Q10. What is a sample?

Answer: A sample is a subset of the population selected for analysis.

Q11. Why do we use a sample instead of the entire population?

Answer: Collecting data from the entire population may be expensive, time-consuming, or practically impossible.

Q12. What is the difference between population mean and sample mean?

Answer: Population mean is the average of the entire population and is represented by μ . Sample mean is the average of a sample and is represented by $\bar{x}$.

Q13. Can sample statistics be used to estimate population parameters?

Answer: Yes. This is one of the main ideas of inferential statistics.

4. Sampling Techniques

Q14. What is sampling?

Answer: Sampling is the process of selecting a subset of observations from a population.

Q15. What is random sampling?

Answer: Random sampling gives each member of the population an equal or known chance of being selected.

Q16. What is stratified sampling?

Answer: In stratified sampling, the population is divided into meaningful groups or strata, and samples are taken from each group.

Q17. When would you use stratified sampling?

Answer: When important subgroups need to be represented in the sample.

Q18. What is systematic sampling?

Answer: Systematic sampling selects observations at a fixed interval from an ordered population, such as every 10th customer.

Q19. What is convenience sampling?

Answer: Convenience sampling selects individuals who are easiest to access. It is simple but may introduce bias.

Q20. Which sampling technique is more likely to create bias: random sampling or convenience sampling?

Answer: Convenience sampling is more likely to create bias because the sample may not properly represent the population.

5. Types of Data

Q21. What is numerical data?

Answer: Numerical data represents quantities that can be measured or counted. Examples: age, salary, sales.

Q22. What is categorical data?

Answer: Categorical data represents groups or categories rather than numerical measurements. Examples: city, gender, product category.

Q23. What is discrete data?

Answer: Discrete data consists of countable values, usually whole numbers. Example: number of customers.

Q24. What is continuous data?

Answer: Continuous data can take any value within a range. Example: height, weight, temperature.

Q25. Is age discrete or continuous?

Answer: Conceptually, age is continuous because it can be measured to arbitrary precision, although it may sometimes be recorded as whole numbers.

6. Scales of Measurement

Q26. What are the four scales of measurement?

Answer: Nominal, Ordinal, Interval, and Ratio.

Q27. What is nominal data?

Answer: Nominal data consists of categories with no natural order. Example: city or blood group.

Q28. What is ordinal data?

Answer: Ordinal data has categories with a meaningful order, but the differences between categories are not necessarily equal. Example: satisfaction level.

Q29. What is interval data?

Answer: Interval data has ordered values with equal intervals, but it does not have a meaningful zero. Example: temperature in Celsius.

Q30. What is ratio data?

Answer: Ratio data has equal intervals and a meaningful zero. Example: salary, weight, height.

Q31. Why does the scale of measurement matter?

Answer: Because it determines which mathematical and statistical operations are meaningful for the data.

7. Mean, Median, Mode

Q32. What is the mean?

Answer: The mean is the sum of all observations divided by the number of observations.

Q33. What is the median?

Answer: The median is the middle value after sorting the data.

Q34. What is the mode?

Answer: The mode is the most frequently occurring value.

Q35. Which measure of central tendency is most affected by outliers?

Answer: Mean.

Q36. Which is usually more robust to outliers: mean or median?

Answer: Median.

Q37. When would you prefer median over mean?

Answer: When the data contains strong outliers or is highly skewed.

Q38. Example: salaries are 20k, 22k, 24k, 25k, 5 lakh. Which is better to represent a typical salary?

Answer: Median, because the very high salary will strongly affect the mean.

8. Measures of Dispersion

Q39. What is dispersion?

Answer: Dispersion describes how spread out the observations are.

Q40. What is range?

Answer: Range = maximum value minus minimum value.

Q41. What is variance?

Answer: Variance measures the average squared deviation of observations from the mean.

Q42. What is standard deviation?

Answer: Standard deviation measures the spread of data around the mean and is the square root of variance.

Q43. Which is easier to interpret in the original unit: variance or standard deviation?

Answer: Standard deviation, because it has the same unit as the original data.

Q44. If standard deviation is very high, what does that suggest?

Answer: The values are more spread out from the mean.

9. Why Sample Variance Uses $n-1$

Q45. Why is sample variance divided by $n-1$ instead of n ?

Answer: Using n tends to underestimate population variance when estimating it from a sample. Dividing by $n-1$, known as Bessel's correction, gives an unbiased estimator under the usual assumptions.

Q46. What is the name of this correction?

Answer: Bessel's correction.

10. Random Variables

Q47. What is a random variable?

Answer: A random variable is a variable that represents the numerical outcome of a random experiment.

Q48. Give a simple example.

Answer: For a coin toss, we can define $X = 1$ for heads and $X = 0$ for tails.

Q49. What is a discrete random variable?

Answer: A random variable that takes countable values. Example: number of customers arriving.

Q50. What is a continuous random variable?

Answer: A random variable that can take any value within a range. Example: temperature.

11. Percentiles and Quartiles

Q51. What is a percentile?

Answer: A percentile tells us the value below which a certain percentage of observations fall.

Q52. What does the 50th percentile represent?

Answer: The median.

Q53. What does the 25th percentile represent?

Answer: The first quartile, Q1.

Q54. What does the 75th percentile represent?

Answer: The third quartile, Q3.

Q55. What is the Interquartile Range?

Answer: $IQR = Q3 - Q1$.

Q56. Why is IQR useful?

Answer: It measures the spread of the middle 50% of the data and is useful for identifying outliers.

12. Five Number Summary

Q57. What are the five numbers in a five-number summary?

Answer: Minimum, Q1, Median, Q3, and Maximum.

Q58. Where is five-number summary commonly visualized?

Answer: In a box plot.

13. Histogram and Skewness

Q59. What is a histogram?

Answer: A histogram is a graph that shows the distribution of numerical data using bins.

Q60. What is skewness?

Answer: Skewness describes the asymmetry of a distribution.

Q61. What are the main types of skewness?

Answer: Symmetric, right/positive skew, and left/negative skew.

Q62. In a right-skewed distribution, which is generally larger: mean or median?

Answer: Mean is generally greater than median.

Q63. Give a practical example of right-skewed data.

Answer: Income or salary data is often right-skewed because a small number of people may have extremely high incomes.

14. Covariance

Q64. What does covariance tell us?

Answer: Covariance indicates the direction in which two variables tend to move together.

Q65. What does positive covariance mean?

Answer: When one variable tends to increase, the other also tends to increase.

Q66. What does negative covariance mean?

Answer: When one variable tends to increase, the other tends to decrease.

Q67. Does covariance tell us the strength of relationship in an easily comparable scale?

Answer: Not reliably, because its magnitude depends on the units of the variables.

15. Correlation

Q68. What is correlation?

Answer: Correlation measures the strength and direction of a linear relationship between two variables.

Q69. What is the range of correlation?

Answer: $-1 \leq r \leq 1$.

Q70. What does correlation +1 mean?

Answer: A perfect positive linear relationship.

Q71. What does correlation -1 mean?

Answer: A perfect negative linear relationship.

Q72. What does correlation close to 0 mean?

Answer: There is little or no linear relationship.

Q73. Does correlation imply causation?

Answer: No. Correlation does not prove that one variable causes the other.

Q74. Give an example where correlation does not imply causation.

Answer: Ice cream sales and swimming incidents may increase together during summer. The common factor is hot weather; one does not necessarily cause the other.

16. Scenario-Based Interview Questions

Q75. Your dataset contains salary values with a few extremely large values. Mean or median?

Answer: Median, because it is less affected by extreme outliers.

Q76. You have customer ages and want to understand their distribution. What would you use?

Answer: A histogram.

Q77. You want to detect outliers using quartiles. What would you calculate?

Answer: Q1, Q3, and IQR.

Q78. Two variables have correlation 0.95. Does that mean one causes the other?

Answer: No. It indicates a strong positive linear relationship, but correlation alone does not establish causation.

Q79. You survey 500 customers out of 1 million customers. What are the 500 customers?

Answer: The sample.

Q80. What is the 1 million customers?

Answer: The population.

Q81. A dataset has mean = 50 and standard deviation = 5. A value is 65. What is its z-score?

Answer: $z = (65 - 50) / 5 = 3$. The value is 3 standard deviations above the mean.

Q82. Which is more robust to outliers: standard deviation or IQR?

Answer: IQR is more robust to outliers.

Q83. Your data is strongly right-skewed. Would mean necessarily be the best measure of central tendency?

Answer: Not necessarily. Median is often more representative.

Q84. If two variables have covariance 0, does that necessarily mean there is no relationship?

Answer: No. It indicates no linear co-movement under the covariance measure; nonlinear relationships can still exist.

17. Quick Numerical Practice

Q85. For data 10, 20, 30, what is the mean?

Answer: 20.

Q86. For data 2, 4, 6, 8, 100, which central tendency measure better represents a typical value?

Answer: Median, because 100 is an outlier.

Q87. For data 1, 2, 3, 4, 5, what is the range?

Answer: $5 - 1 = 4$.

Q88. If $Q1 = 20$ and $Q3 = 50$, what is IQR?

Answer: $50 - 20 = 30$.

Q89. If a correlation coefficient is -0.85, how would you describe the relationship?

Answer: A strong negative linear relationship.

Q90. If a correlation coefficient is 0.05, what can you conclude?

Answer: There is little or no linear relationship. It does not prove that no relationship of any kind exists.

Final Interview Checklist

- Explain mean, median, mode without formulas first.
- Know why median is useful with outliers and skewed data.
- Explain variance vs standard deviation and the role of $n-1$.
- Calculate Q1, Q3, IQR and explain outlier detection.
- Read a histogram and identify skewness.
- Explain covariance vs correlation.
- Know that correlation does not imply causation.
- Distinguish population, sample, statistic, and parameter.
- Explain common sampling techniques and their trade-offs.
- Use small numericals and Python/NumPy/Pandas to verify results.

Formula Quick Sheet

Mean	sum of values / number of values
Range	max - min
IQR	Q3 - Q1
Sample variance	sum of squared deviations / ($n - 1$)
Z-score	$(x - \text{mean}) / \text{standard deviation}$
Correlation range	-1 to +1